

Reflective Journal

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Class: 6263-ITAI-1371 Intro to Machine Learning-S10-12321

Module 06 Lab — Regression and Classification (Team ML_1371_12321)

In this lab, we focused on building, training, and evaluating machine learning models using the cleaned Titanic dataset. Our primary goal was to implement both a regression pipeline to predict continuous values (Fare) and a classification pipeline to predict discrete labels (Survived), while measuring how well each model performed using appropriate metrics.

First, we built a LinearRegression model to predict ticket fares based on passenger age and class. We trained the model on our training set, generated predictions on the test set, and evaluated the results using mean_squared_error (MSE). Next, we trained a LogisticRegression model to predict passenger survival and evaluated its performance using accuracy_score. Finally, we inspected model parameters like .coef_ to understand how individual features influenced our model's predictions.

What we found most challenging initially was understanding how different metrics apply to different machine learning tasks specifically why accuracy cannot evaluate a regression model where continuous outputs rarely match exact floating-point numbers. What clicked for us was seeing how scikit-learn simplifies the model training workflow: calling .fit(), .predict(), and an evaluation function creates a complete machine learning pipeline in just a few lines of code. We can easily see how this connects to real-world applications, such as financial institutions predicting house prices with regression models or medical tools classifying patient diagnostics using classification models.